

Matrix Algebra

M514: Econometrics II

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Matrix multiplication

Let A , and B be $n \times m$, $m \times p$ dimensional matrices given by

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{pmatrix}, B = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1p} \\ b_{21} & b_{22} & \dots & b_{2p} \\ \vdots & & \dots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mp} \end{pmatrix}$$

The (i, j) -th element of AB , i.e., i -th row and j -th column of the matrix multiplication of A and B is given by

$$[AB]_{ij} := \sum_{k=1}^m a_{ik} b_{kj}$$

Each element of AB is given by sum over m scalars.

Matrix multiplication

Let $A_{\bullet j}$ be the j -th column of A and $A_{i\bullet}$ be i -th row of A . Then, we have

$$AB = \sum_{k=1}^m A_{\bullet k} B_{k\bullet}$$

In other words, matrix multiplication is given by sum over m rank 1 matrices. Also,

$$[AB]_{i\bullet} = A_{i\bullet} B, [AB]_{\bullet j} = AB_{\bullet j}$$

In other words, in calculating the i -th row of AB , only i -row of A is needed. In calculating the j -th column of AB , only j -th column of B is needed. This means that we have

$$AB = \begin{pmatrix} AB_{\bullet 1} & \dots & AB_{\bullet m} \end{pmatrix} = \begin{pmatrix} A_{1\bullet} B \\ \dots \\ A_{n\bullet} B \end{pmatrix}$$

Matrix multiplication

Let $n \times k$ dimensional observation matrix X be given by

$$X = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}$$

where we may interpret each $x_i \in \mathbb{R}^k$ as observation (sample). Then, $X'X = \sum_{i=1}^n x_i x'_i$. Let $n \times 1$ dimensional vector Y be given by $y = (y_1, \dots, y_n)'$. Then, we have $X'y = \sum_{i=1}^n x_i y_i \in \mathbb{R}^k$. (Memorize this!)

$$\begin{aligned} \hat{\beta} &= (X'X)^{-1} X'y \\ &= \left(\sum_{i=1}^n x_i x'_i \right)^{-1} \left(\sum_{i=1}^n x_i y_i \right) \end{aligned}$$

Linear independence

$x_1, x_2, \dots, x_m \in \mathbb{R}^n$ are said to be *linearly independent* if

$$a_1x_1 + a_2x_2 + \dots + a_mx_m = 0 \iff a_1 = a_2 = \dots = a_m = 0$$

where $a_1, \dots, a_m \in \mathbb{R}$. Note that the zero on the left is an n dimensional vector of zeros, whereas the zero on the right is a scalar.

$x_1, x_2, \dots, x_m \in \mathbb{R}^n$ are said to be *linearly independent* if for any $i \in \{1, \dots, m\}$ and for any $a_j, j = 1, \dots, i-1, i+1, \dots, m$, we cannot have

$$x_i = a_1x_1 + \dots + a_{i-1}x_{i-1} + a_{i+1}x_{i+1} + \dots + a_mx_m$$

If $m > n$, $x_1, x_2, \dots, x_m \in \mathbb{R}^n$ cannot be linearly independent.

Dimension of a vector space V

A *vector space* is a set V with addition and scalar multiplication on V such that for all $u, v \in V$ and $a, b \in \mathbb{R}$ we have

$$au + bv \in V$$

A *basis* of V is a collection of vectors that is linearly independent and spans V . We say $(v_1, \dots, v_m)$ spans V if any element of V can be written as linear combination of $(v_1, \dots, v_m)$. For any collection $v_1, \dots, v_m \in V$, the span is defined as

$$\text{span}(v_1, \dots, v_m) = \left\{ y \in V : y = \sum_{i=1}^m a_i v_i, a_1, \dots, a_m \in \mathbb{R} \right\}$$

The dimension of V is the number of vectors in a basis of V .

Range of A

Let A be $n \times m$.

$$R(A) := \{y \in \mathbb{R}^n : y = Ax, x \in \mathbb{R}^m\}$$

Note that

$$y = Ax = \sum_{i=1}^m A_{\bullet i} x_i$$

is a linear combination of columns of A . This means that the range of A is the column space of A . The rank of A is defined as the dimension of the range of A , written as

$$\text{rank}(A) = \dim(R(A))$$

This is equal to the number of linearly independent columns of A . A is said to be *full column rank* if $\text{rank}(A) = m$, i.e., all columns of A are linearly independent.

Null space of A

Let A be $n \times m$.

$$N(A) = \{x \in \mathbb{R}^m : Ax = 0\}$$

If A is of full column rank, $\dim(N(A)) = 0$, i.e., only $x = 0$ exists in $N(A)$. More generally, we have

$$\text{rank}(A) + \dim(N(A)) = m$$

This fundamental result is called the rank–nullity theorem.¹

¹We won't prove this in this class.

Eigenvectors

Consider an $n \times n$ dimensional matrix A . $\lambda \in \mathcal{C}$ is an eigenvalue of A if there exists a non-zero $v \in \mathbb{R}^n$ s.t.

$$Av = \lambda v$$

Suppose such a v exists. Then, we have

$$(A - \lambda I_n)v = 0$$

If the inverse of $(A - \lambda I_n)$ exists, we can multiply each side with the inverse to get $v = 0$. Thus, for a non-zero v to exist, we need $\det(A - \lambda I_n) = 0$. Note that $\det(A - \lambda I_n)$ is n -th order polynomial of λ . Thus eigenvalues are the solution to the n -th order equation $\det(A - \lambda I_n) = 0$. This is one of the defining properties of eigenvalues. By the fundamental theorem of algebra, we have n solutions for λ . The solutions λ^* are generally complex and possibly repeated.

Eigenvectors: Example 1

Find eigenpairs of

$$A = \begin{pmatrix} -5 & 2 \\ -9 & 6 \end{pmatrix}$$

Eigenvectors: Example 2

Suppose A is a diagonal matrix given by

$$A = \begin{pmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{pmatrix}$$

Then, we have

$$A - \lambda I_n = \begin{pmatrix} a_{11} - \lambda & & & \\ & a_{22} - \lambda & & \\ & & \ddots & \\ & & & a_{nn} - \lambda \end{pmatrix}$$

and $\det(A - \lambda I_n) = (a_{11} - \lambda) \cdots (a_{nn} - \lambda)$. Thus, eigenvalues of A are given by the diagonal elements of A .²

²**FACT:** Determinant of a diagonal matrix is the product of diagonal elements.

Eigenvectors: Example 2

Let λ_i be the i -th eigenvalue of A . Then, we have

$$\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i \quad (1)$$

$$\det(A) = \prod_{i=1}^n \lambda_i \quad (2)$$

(1) and (2) hold not only for diagonal matrices but for any square matrix A .

Eigenvectors

1. If λ is an eigenvalue of A , then the subspace $\{v : Av = \lambda v\}$ is called the eigenspace of A associated with λ . The eigenspace is in general different from the column space of A .
2. Eigenspace associated with zero eigenvalues of A is the null space of A .
3. The span of the eigenvectors with non-zero eigenvalues is contained in the column space.

Direct sum of vector spaces

Let X and Y be nonempty subsets of a vector space V . The sum of X and Y is defined as

$$X + Y = \{x + y : x \in X, y \in Y\}$$

Suppose further that X and Y are subspaces of a vector space V . We say V is the direct sum of X and Y and write $V = X \oplus Y$ if every vector $v \in V$ can be expressed *uniquely* as the sum $v = x + y$ where $x \in X$ and $y \in Y$.

FACT:

If $V = X + Y$, V is direct sum of X and Y if and only if $X \cap Y = \{0\}$.³

³We won't prove this in this class.

Projection

Let P be $n \times n$ dimensional matrix. P is called *idempotent* if

$$P^2 = P$$

All eigenvalues of an idempotent matrix are either 1 or 0. To see this, suppose λ is an eigenvalue of P . Then, we have

$$\begin{aligned} Pv &= \lambda v \\ \implies \lambda v &= Pv = P^2v = P(Pv) = P(\lambda v) = \lambda(Pv) = \lambda^2 v \end{aligned}$$

Since v is non-zero, we must have $\lambda = 0$ or 1 . If P is an idempotent matrix, $I - P$ is also an idempotent matrix. The only invertible idempotent matrix is the identity matrix.

Projection

For any $x \in \mathbb{R}^n$, we may write

$$x = Px + (I_n - P)x \tag{3}$$

1. $Px \in R(P)$ by definition.
2. $(I_n - P)x \in N(P)$ because $P(I_n - P)x = (P - P)x = 0$.

3. **CLAIM:**

Decomposition of the form (3) is unique in the sense that if x admits a representation $x = u + v$ where $u \in R(P)$ and $v \in N(P)$, we must have $u = Px$ and $v = (I_n - P)x$.

Projection

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Proof:

Using the **FACT** two slides back, ISTS: $R(P) \cap N(P) = \{0\}$. We first show that $R(I - P) = N(P)$, and then show $R(P) \cap R(I - P) = \{0\}$

1. $R(I - P) \subset N(P)$ because any $(I_n - P)x \in N(P)$.
2. $N(P) \subset R(I_n - P)$ because for any x such that $Px = 0$, we have $x = Px + (I_n - P)x = (I_n - P)x$.
3. We have $R(P) \cap R(I - P) = \{0\}$. To see this, suppose $x \in R(P) \cap R(I - P)$. This means that there exists u, v such that $x = Pu = (I - P)v$. We have $x = Pu = P(Pu) = Px = P(I - P)v = 0$, as desired.

Projection

1. Since x admits a unique representation given by $x = u + v, u \in R(P), v \in N(P)$ with $Px = Pu = u$, the linear transformation given by an idempotent matrix P can be regarded as the projection onto $R(P)$ along $N(P)$.⁴ Thus, an idempotent matrix is also called a projection.
2. If the idempotent matrix P is also symmetric, P is called orthogonal projection. This is because the decomposition $x = Px + (I - P)x$ is orthogonal. To see this, note that $(Px)'(I - P)x = x'P'(I - P)x = x'P(I - P)x = 0$. Thus, a symmetric idempotent matrix is also called (orthogonal) projection.⁵

⁴Let $P = \begin{pmatrix} 3 & -6 \\ 1 & -2 \end{pmatrix}$. What is the range and null space of P ?

⁵Show range and null space of an orthogonal idempotent matrix is orthogonal.

Projection: Example

$$P = \begin{pmatrix} 3 & -6 \\ 1 & -2 \end{pmatrix}$$

1. Draw the range of P in a 2-dimensional plane.
2. Draw the null space of P in a 2-dimensional plane.
3. Project a vector $(5, 2)'$ onto range of P along the null space of P

Projection

Let A be $n \times m$ matrix with full column rank. Let B be $n \times m$ dimensional matrix such that $B'A$ is non-singular.⁶ Then, projection onto column space of A along null space of B' is defined as

$$A(B'A)^{-1}B'$$

Orthogonal projection onto the column space of A is defined as

$$A(A'A)^{-1}A'$$

Orthogonal projection onto the null space of A is given by

$$I - A(A'A)^{-1}A'$$

This is because $I - A(A'A)^{-1}A'$ is orthogonal idempotent matrix, and because range of $A(A'A)^{-1}A'$ is equal to the null space of $I - A(A'A)^{-1}A'$.

⁶I.e., inverse exists.

Projection: Exercise

Let A be $n \times m_1$ and B be $n \times m_2$ dimensional matrices. If A and B are orthogonal, i.e., $A'B = 0$, orthogonal projection jointly onto column spaces of A and B is equal to the sum of the projection onto column space of A and the projection onto the column space of B .

Projection

CLAIM:

Orthogonal projection of x to column space of A gives us an element in the range of A such that the distance between the projection and x is minimized.

Proof:

To see this, note that for all $y \in R(A)$, we have

$$\|x - y\|^2 = \|x - P_A x\|^2 + \|P_A x - y\|^2 \geq \|x - P_A x\|^2$$

by the Pythagoras theorem and non-negativity of norms. More concretely, consider the following problem:

$$\min_{y \in \mathbb{R}^m} \|Ay - x\|^2 = (Ay - x)'(Ay - x)$$

The first order condition gives us the solution $y^* = (A'A)^{-1}A'x$, as desired.

Spectral decomposition

FACT: Let A be $n \times n$ dimensional symmetric matrix. Then, the following hold:

- ▶ All eigenvalues are real.
- ▶ Eigenspaces associated with distinct eigenvalues are orthogonal.
- ▶ Eigenvectors of A span $\mathbb{R}^n$.

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WLOG, suppose $(\lambda_1, v_1), \dots, (\lambda_n, v_n)$ are eigen-pairs of A . Since the span of $(v_1, \dots, v_n)$ is equal to $\mathbb{R}^n$, projection onto $(v_1, \dots, v_n)$ is an identity matrix, i.e., $\sum_{i=1}^n v_i(v_i'v_i)^{-1}v_i' = I_n$. Thus, any $x \in \mathbb{R}^n$ can be written as

$$x = \sum_{i=1}^n v_i(v_i'v_i)^{-1}v_i'x = \sum_{i=1}^n \left(\frac{v_i'x}{v_i'v_i} \right) v_i$$

⁷We won't prove this in class.

Spectral decomposition

This means that Ax can be written as

$$\begin{aligned} Ax &= \sum_{i=1}^n \left(\frac{v_i' x}{v_i' v_i} \right) A v_i \\ &= \sum_{i=1}^n \left(\frac{v_i' x}{v_i' v_i} \right) \lambda_i v_i \\ &= \sum_{i=1}^n \lambda_i v_i (v_i' v_i)^{-1} v_i' x \end{aligned}$$

Since the above holds for all $x \in \mathbb{R}^n$, we have $A = \sum_{i=1}^n \lambda_i v_i (v_i' v_i)^{-1} v_i'$. WLOG, suppose that we normalize the eigenvectors so that $v_i' v_i = 1$. Then, we have

$$A = \sum_{i=1}^n \lambda_i v_i v_i' \tag{4}$$

(4) is called the spectral representation of A .

Spectral decomposition

Let U and D be given by

$$U = (v_1 \quad \dots \quad v_n), D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

Then, we have

$$A = UDU'$$

Definite matrix

1. A symmetric matrix A is called positive (negative) definite if all its eigenvalues are positive (negative).
2. A symmetric matrix A is called positive (negative) semi-definite if all its eigenvalues are non-negative (non-positive).

For any nonzero $x \in \mathbb{R}^n$, we may write

$$\begin{aligned}x'Ax &= x' \left(\sum_{i=1}^n \lambda_i v_i v_i' \right) x \\ &= \sum_{i=1}^n \lambda_i (x' v_i)^2\end{aligned}$$

Thus, A is positive definite if and only if $x'Ax > 0$ for all non-zero $x \in \mathbb{R}^n$ and A is positive semi-definite if $x'Ax \geq 0$ for all $x \in \mathbb{R}^n$. Likewise for negative (semi-)definite matrix A .